

Chapter 7:Radiochemical techniques

Introduction

- Atoms with the **same number of protons** but a **different number of neutrons** are called **isotopes**.
- To identify an **isotope** we use the symbol ${}^A_Z E$
 - **Where**
 - **E** is the element's **atomic symbol**,
 - **Z** is the element's **atomic number** (*which is the number of protons*), and
 - **A** is the element's **atomic mass number** (which is the sum of the *number of protons and neutrons*)

- Although isotopes of a given element have the same **chemical properties**, their **nuclear properties are different**.
- The most important difference between isotopes is their **stability**.
- The nuclear configuration of **a stable isotope remains constant with time**.
- Unstable isotopes, however, **spontaneously disintegrate**, **emitting radioactive particles** as they transform into **a more stable form**.

Types of radiation

- **Three types of radiation;**
 - **α -particles,**
 - **β -particles, and**
 - **γ -rays.**

- **Radioactive decay** is a random 1st order process independent on the temperature & chemical state of the radionuclide
- The fundamental radioactive decay law is: -
 - $A = A_0 e^{-\lambda t}$
- Where; A is activity at time t; A_0 is initial activity; & λ is the decay constant.
- Half-life of a radionuclide is defined as the time required for the activity to decay to half of its initial value

- Radioisotopes with comparatively short half-lives, measured in **hours** or **days**, are widely used in radiopharmaceutical preparations
- Used as diagnostic agents & in the treatment of neoplastic diseases
- These include ^{32}P , ^{51}Cr , ^{57}Co , ^{59}Fe , ^{75}Se , $^{99\text{m}}\text{Tc}$, ^{125}I , ^{131}I , & ^{133}Xe
- There are different requirements for a radiopharmaceutical; as shown in the next-coming slides

1. Properties of the radionuclide

a. Suitable half-life

- A few hours of half-life is needed

b. Pure gamma (γ) emitter of suitable energy

- There should be no α & β emissions
- The γ energy should be high-enough for the radiation to penetrate out from internal organs
- The range 100-200 KeV is optimal

c. Capable of being produced daily

- The radionuclide must be suitable for daily production in the hospital pharmacy
- Because it can't be supplied from any distance

d. It must be capable of being converted to a range of chemical entities for imaging a range of organs

- The radionuclide ^{99m}Tc fulfils all of the above criteria & is the radionuclide of choice for most radiopharmaceuticals.

2. **Pharmaceutical properties**

- Normal requirements for an injection must be applied.

3. **Chemical properties**

- The correct chemical (or physicochemical) form is needed to target to the required organ
- Pharmacologic & toxicologic aspects are also important but, due to extremely small masses injected, they're generally not a problem

❖ **Analytical utility** of radiopharmaceuticals include their use as **tracers** in **analysis** & in **distribution & metabolism** studies of drugs in animals

Different types of radiation

i. Alpha particles

- They are Helium nuclei (${}^4\text{He}^{2+}$) having a mass of 4 & charge +2
- Have low penetrating power, being stopped by 1-2cm air, & by a sheet of white paper
- They cause intense ionization, 10^4 - 10^5 ion pairs per cm in air

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- **An alpha particle**, which is symbolized as **α** , is equivalent to a helium nucleus, ${}^4_2\text{He}$
- Thus, **emission of an alpha particle** results in a **new isotope** whose **atomic number** and **atomic mass number** are, respectively, **2 and 4** less than that for the **unstable parent isotope**.
- Example ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \alpha$

ii. Beta particles

- High-energy, **high-speed electrons or positrons** emitted by certain types of radioactive nuclei such as ^{40}K
- Can penetrate up to 50cm in air & up to 0.5cm in aluminum.
- Ionization in air is 10^2 - 10^3 ion pairs/cm
- From a collection of identical nuclei (e.g. ^{14}C) β particles with a range of energies are emitted

- Produced when a neutron converts to a proton, or a proton converts to a neutron.
- **A negatron**, is produced when a **neutron is converted to a proton**, increasing the **atomic number by 1**. (${}_{-1}^0\beta$)
 - Example: ${}_{82}^{214}\text{Pb} \rightarrow {}_{83}^{214}\text{Bi} + {}_{-1}^0\beta$
- **Positron**: The beta particle formed when a **proton converts to a neutron** (${}_{+1}^0\beta$).
 - Example ${}_{15}^{30}\text{P} \rightarrow {}_{14}^{30}\text{Si} + {}_{+1}^0\beta$

iii. Gamma radiation

- Electromagnetic radiation similar to light & X-rays but of higher energy (up to 3 MeV) & shorter wavelength
- Very penetrating
- Low ionizing power (the order of 1-10 ion pairs/cm)
- Unlike β -particles (β -radiation), all γ -rays from a collection of identical nuclei have the same energy

Measurement of radioactivity

- 3 general categories of methods for detecting & measuring radiation, depending on the particular type of radiation property utilized:
 1. Property of radiation to cause ionization (in a gas)
 2. Property of radiation to cause scintillation (fluorescence)
 3. Property of radiation to effect chemical changes

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a. **Geiger-Muller counting**

- 'Counting' implies combination of detection & display on a scaler or rate meter.
- A method for counting β particles.
- It's based on the ability of radiation to cause ionization in a gas
- Easily ionizable gases like argon are used
- A β particle enters the Geiger tube & causes some argon atoms to become ionized
- These ions attracted to the outer negative electrode by the potential gradient present.

- Further ions will be formed & in turn attracted to the anode
- Further ionization occurs until the whole volume of the detector tube is ionized

b. Liquid scintillation counting

- Great improvement in measurement of low energy **β -emitters** like ^3H & ^{14}C which have wide application in tracer work.
- Sample is dissolved in a solvent (e.g. toluene or dioxane) together with some scintillant material, such as diphenyloxazole

- Scintillation implies the whole process of events whereby energy of emitted β -particle is transferred via solvent to scintillant molecules
- Light emission by these molecules will then be detected by photomultipliers
- Due to very low light intensity involved, photomultipliers need to be operated at high sensitivity
- This results in an increase in electronic noise

- To reduce this background noise, 2 photomultipliers are used in 'coincidence counting' mode
- This way, only output pulses coincident in time are counted
- Random noise pulses created independently in each photomultiplier will not, in general, be detected

NB. A solid scintillation process is used in the measurement of γ -radiation.

Radiochemical methods in analysis

a. Isotope Dilution Analysis (IDA)

- Homogeneous dilution of a labelled compound of known specific activity with the same unlabelled compound
- Its application results from the fact that the compound need not be isolated quantitatively
- It's necessary only to isolate a pure sample of the compound
- Two forms of this method available

1. Direct IDA

- A known weight of labelled cpd of **known specific activity** (activity/unit mass) is added to a mixture containing an unknown amount of unlabelled cpd
- If,
 - A_o = original activity of added labelled cpd,
 - M_o = original amount of added labelled cpd, &
 - S_o = original specific activity of added labelled cpd
- Then, **$S_o = A_o / M_o$**

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- Let, M_u = amount of unlabelled cpd present (unknown) & S_i = specific activity of the final isolated sample. Then,
 - $S_i = A_o / (M_o + M_u)$ (assume whole sample isolated)
 - Substituting for A_o ($A_o = S_o M_o$ from above)
 - $S_i = S_o M_o / (M_o + M_u)$
- Therefore; $M_u = [(S_o / S_i) - 1] M_o$

2. Inverse or reverse IDA

- Known quantity of non-labelled compound is added to a mixture containing an unknown amount of labelled compound

b. Radioimmunoassay (RIA)

- Based on competition between the labelled & unlabelled cpd for binding to a specific antiserum/antibody prepared against the cpd.

- Proportion of labelled cpd bound to the antibody is related to the concentration of unlabelled cpd (in the sample)
- Highly sensitive technique
- Enables direct measurement of antigen concentration (not biological response)

NB. – Antibody should not be able to distinguish between labelled & unlabelled antigen

- No. of labelled + unlabelled antigen should be $>$ No. of binding sites on antibody

GOOD MANUFACTURING PRACTICE (GMP) FOR RPS

- Should be sterile, pyrogen-free and safe.
- Rp production should be undertaken in a way compliant with the legal framework and regulatory procedures and in accordance with the principles of GMP for medicinal products.
- GMP requires dedicated clean room facilities with the highest classification to ensure aseptic preparation of rps.
- Preparation and quality control of rps should be conducted only by competent and appropriately qualified personnel.

✓ **The label on the primary container should include:**

- A statement that the product is radioactive or the international symbol for radioactivity
- The name of the radiopharmaceutical preparation
- The route of administration;
- The expiry date and
- The batch (lot) number assigned by the manufacturer;
- For solutions, the total volume.



The trefoil symbol used to indicate ionising radiation.



2007 ISO radioactivity danger symbol intended for IAEA Category 1, 2 and 3 sources defined as dangerous sources capable of death or serious injury



The dangerous goods transport classification sign for radioactive materials

Storage of Radioactive Substances

- Radiopharmaceuticals should be kept in well-closed containers.
- Care should be taken to comply with national regulations for protection against ionizing radiation.
- The storage and distribution of all radioactive products should be controlled to minimise any risks with respect to their quality.

Quality control

- **Quality control should cover the sampling and testing of both**
 - The starting materials and
 - The radiopharmaceutical final drug products
- Ensure that materials are not released for use until their quality has been determined to conform to the predefined acceptance specifications.

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- The finished products should contain active ingredients that are compliant with the qualitative and quantitative composition stipulated by the marketing authorization.

